

Skills Summary

Converting Between Exponential and Logarithmic Forms

Use this reference while completing your worksheet or when studying.

A logarithm is an exponent. The two forms hold the same three numbers:

$$b^y = x \quad \Longleftrightarrow \quad \log_b x = y$$

- b is the **base** — the subscript in log form, the bottom number in exponential form. It never changes role.
 - y is the **exponent** — and it is what the logarithm equals.
 - x is the **result** — and it is what the logarithm is taken of. Only positive x is allowed.
- Choose the form that puts your unknown alone: unknown exponent $\rightarrow$ log form; unknown inside the log $\rightarrow$ exponential form.

1. Exponential to logarithmic

Label base, exponent, result first. Then write $\log_{\text{base}}(\text{result}) = \text{exponent}$. The base drops to the subscript; nothing else moves.

Example: Rewrite $4^3 = 64$ in logarithmic form and give the value.

Step 1. Name the roles before rearranging: base 4, exponent 3, result 64.

Step 2. The base becomes the subscript and the exponent becomes the value, so $\log_4 64 = 3$.

$$\log_4 64 = 3$$

Try it: Rewrite $7^2 = 49$ in logarithmic form and give the value of $\log_7 49$.

check: 2

2. Logarithmic to exponential

$\log_b x = y$ rebuilds as $b^y = x$: the subscript is the base, the value of the logarithm is the exponent. Use this direction whenever the unknown sits inside the logarithm.

Example: Solve $\log_6 x = 3$.

Step 1. The unknown is what the logarithm is taken of, so exponential form is the one that isolates it.

Step 2. $\log_6 x = 3$ becomes $x = 6^3$, and $6^3 = 216$.

$$x = 216$$

Try it: Solve $\log_9 x = \frac{1}{2}$.

check: 3

3. Evaluating a logarithm by converting

To evaluate $\log_b x$, set it equal to y , convert to $b^y = x$, and write both sides as powers of the same number. A result below 1 forces a negative exponent; a result between 1 and the base forces a fraction between 0 and 1.

Example: Evaluate $\log_2 \frac{1}{16}$.

Step 1. Convert to a power question, then write both sides on the same base so the exponents can be compared directly.

Step 2. $2^y = \frac{1}{16} = 2^{-4}$, so $y = -4$.

$$\log_2 \frac{1}{16} = -4$$

Try it: Evaluate $\log_{25} 5$ exactly.

check: 1/2

4. Solving when the unknown is the exponent

$b^x = k$ converts to $x = \log_b k$. For a decimal, change base: $\log_b k = \frac{\ln k}{\ln b}$ — a quotient of two logarithms, never the logarithm of a quotient. Bracket the answer with nearby whole powers first.

Example: Solve $3^x = 50$ to three decimals.

Step 1. The unknown is the exponent, so convert to logarithmic form rather than dividing.

Step 2. $x = \log_3 50 = \frac{\ln 50}{\ln 3}$, and since $3^3 = 27$ and $3^4 = 81$, the answer must sit between 3 and 4.

$$x = 3.561$$

Try it: Solve $5^x = 90$ to three decimals.

check: 2.792

(!) **Watch out:** the subscript is always the base — $\log_2 32 = 5$ rebuilds as 2^5 , never 5^2 . And $\frac{\ln 20}{\ln 3}$ is not $\ln\left(\frac{20}{3}\right)$. Finally, $\log_b x$ exists only for positive x , so check the inside of every logarithm after solving.